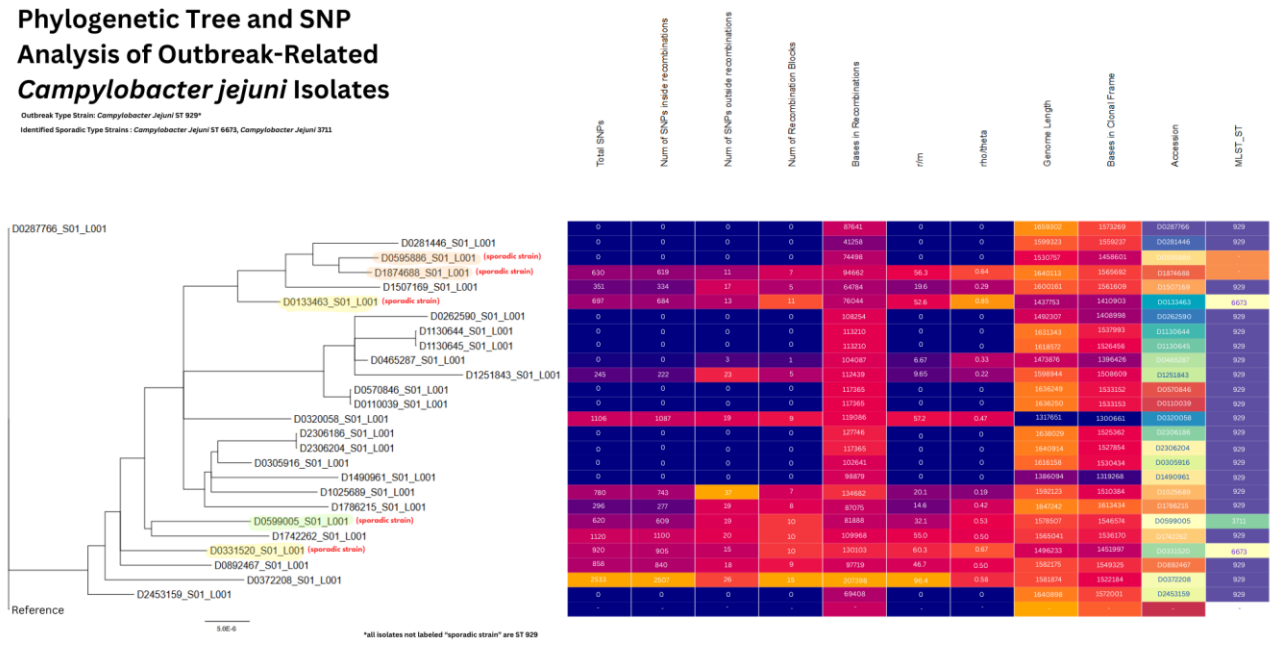
Public Health Recommendations to the CDC

Summary:
This report investigates a *Campylobacter jejuni* outbreak through phylogenetic and MLST analysis, identifying both outbreak-associated and sporadic isolates. The primary outbreak strain was determined to be MLST Sequence Type 929, while five isolates with distinct genetic profiles were classified as sporadic. Based on genomic assessments and sample traceability, public health recommendations are provided to mitigate further transmission and prevent future outbreaks.

I. Identification of Outbreak and Sporadic Samples

The outbreak under investigation is attributed to *Campylobacter jejuni*. Analysis of phylogenetic relationships and Multilocus Sequence Typing (MLST) revealed that samples D0331520, D0599005, D0133463, D1874688, and D0595886 are sporadic isolates and do not belong to the main outbreak cluster. These sporadic isolates were identified based on distinct branch lengths, clear separation within the phylogenetic tree, and notably different MLST Sequence Types (STs). The primary outbreak isolates predominantly exhibit MLST ST 929, whereas the sporadic isolates have significantly different STs, including 3711, 6673, or missing data indicated by "-".

The primary outbreak group includes all other analyzed samples, predominantly sourced from milk and food products, indicating a foodborne transmission route. Sample D0287766 served effectively as the in-outbreak reference sample due to superior genomic quality indicators highest N50 value, best auN score, lowest number of contigs, and longest contig lengths as determined through Quast genome assessment.



II. Recommendations for Public Health Actions:

1. Immediate notification should be issued to grocery stores, milk producers, and food distribution centers implicated in the outbreak. These facilities must be instructed to cease distribution of potentially contaminated products and initiate recalls promptly.

2. Public advisories and informational bulletins should be released swiftly to warn consumers about the outbreak, clearly specifying avoidance of milk and possibly related food products sourced from affected regions. It should be emphasized that raw or inadequately pasteurized milk poses significant health risks and should be avoided, particularly in areas linked to the current outbreak.

3. Enhanced surveillance and prompt reporting protocols should be established at healthcare facilities within and surrounding the affected areas. Hospitals and clinics should be alerted to anticipate potential cases, enabling early detection and appropriate clinical management to minimize severe health outcomes.

4. Investigations should be expanded to include environmental sampling, specifically targeting water sources, dairy farms, and food processing facilities, to conclusively determine and mitigate the original contamination source. Particular attention should be paid to agricultural water sources or animal reservoirs that could have facilitated the entry of *Campylobacter jejuni* into the food supply.

5. Continuous genomic monitoring and routine MLST analysis should be implemented for rapid identification of outbreak-related and sporadic strains to allow timely intervention and prevent future outbreaks.

Through coordinated efforts encompassing public advisories, proactive clinical preparedness, and targeted environmental investigations, public health authorities can effectively mitigate the spread and health impact of this *Campylobacter jejuni* outbreak.